

Sam Watts

Machine Learning Engineer

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I'm a data professional with a wide range of experience in different types of modelling and software engineering practices. I originally specialised in recommender systems, but more recently I have followed my interest in other topics, including: geospatial data, computer vision, and agentic systems.

I'm driven by a desire to learn, solve important problems, and be helpful. I'm most interested in climate tech and health, but I'm open to other roles with a positive social impact.

I'm currently based in Edinburgh, Scotland

Right now I'm looking for roles that are in-person in Edinburgh, fully remote, or hybrid in the UK with occasional travel.

Experience

Freelance

Software Engineer • Sep 2026 – Now

Creating ML coding tasks used as RLHF training data for agentic coding models.

GreenFlip

Founding ML Engineer • Sep 2025 – Sep 2026 • 1yr

I built ML models to assess residential homes for low-carbon heating retrofit, working across computer vision, geospatial, and tabular modelling. My solar panel placement algorithm had higher accuracy on maximum placeable panels than the Google Solar API on our internal evaluation set of diverse UK homes.

Continuum Industries

Senior Software Engineer • May 2024 – Aug 2025 • 1yr 4m

I worked with genetic algorithms, machine learning and geospatial data to create tools that speed up the design of large-scale linear infrastructure projects - most notably electricity transmission lines, whose fast development is crucial for the transition to renewable energy.

- Led improvements to genetic algorithms, including smart visualisation of spatially diverse routes and efficient parallel runs of the core algorithms
- Developed 2 ML prototypes to extract data from analogue maps and measure the width of roads from satellite imagery
- Developed LLM-based tools for analysing possible transmission routes and generating technical reports

University of Edinburgh

Research Fellow • Aug 2023 – Mar 2024 • 8m

I worked with a novel retinal image dataset to develop deep learning models for the early detection of age-related macular degeneration, glaucoma and other retinal diseases.

- Implemented semantic segmentation and classification models in PyTorch
- Worked with retinal imaging foundation models using ViT architectures

Too Good To Go

Machine Learning Engineer • Oct 2021 – Mar 2023 • 1yr 6m

I worked on building a product recommendation system from the ground up. My work spanned a range of ML areas, from EDA and model development to CI/CD and productionising model workflows.

- Designed and implemented a recommender model that increased user purchase conversion by 0.6pp versus a baseline in A/B tests (projected value of £1m+ per year)
- Led the adoption of Metaflow as our ML workflow tool, significantly speeding up development
- Experimented with new modelling approaches inspired by research presented at RecSys 2022

John Lewis Partnership

Lead Data Scientist, Product Recommendations • Mar 2021 – Oct 2021 • 8m

Led a team of 4 in the design and execution of data science and data engineering work, focusing on development of existing and new recommendations models - whilst still writing production code. Achievements:

- Implemented hyperparameter tuning using sklearn pipelines across all in-house models - boosting recommendation performance in A/B tests
- Iterated on a sequence learning (Prod2Vec) model built on web browsing data in TensorFlow by exposing extra hyperparameters

Data Scientist • Apr 2018 – Mar 2021 • 2yrs 11m

I joined as one of the first members of a new Data Science team, working across product demand forecasting, supply chain optimisation with vehicle routing models, warehouse simulation, and counterfactual evaluation of experiments. From 2020 I specialised in product recommendations, developing in-house algorithms and making sure they delivered good results in production.

- Productionised a recommendation model on GCP, using AI Platform for training and serving
- Drove implementation of more granular A/B test analysis using Airflow and BigQuery
- Organised a regular journal club, using recent publications to inform our product roadmap

Education

University College London (UCL)

Natural Sciences, BSc - Organic Chemistry and Neuroscience • 2013 – 2017

Projects

Carbon13 Venture Builder Programme

Venture Founder • 2025

I explored multiple areas of climate tech from both a business and technical perspective. With a co-founder, I worked on validating software tooling for recipe formulation in the food industry, with a focus on novel ingredients from precision fermentation and cultivated meat. We ultimately decided not to continue the venture together, but it was a valuable crash course in early-stage company building.

DataKind

DataDive Volunteer & Data Ambassador • 2021 - 2025

I have volunteered on multiple DataDives - which help charities extract value from their data using data science techniques. I also helped to organise several of these events as a Data Ambassador - by framing key analytical questions, cleaning and documenting data, and liaising with charity representatives to enable volunteers to develop good solutions.

Saturdays.AI Copenhagen

Volunteer Mentor • 2022

Helped teach data science & machine learning foundations to a group of beginners. I taught two sessions, on Exploratory Data Analysis and Clustering.

Tools

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| • Python | • Docker |
| • ML libraries - sklearn, tensorflow, pytorch, pandas | • CI/CD - Gitlab, Jenkins, Github Actions |
| • SQL - Redshift, Bigquery, MSSQL, Teradata, dbt | • ML Workflows - Metaflow, AWS Sagemaker |
| • Cloud Platforms - AWS & Google Cloud | • Terraform |
| • Git | • Airflow |